

GeoNail System Architecture

Smart Subterranean Anchoring, Slope Stability Monitoring & Real-Time Digital Twin Platform

Core Domain: IoT & Geotechnical Safety	Deployment: Open-Cast Mines & Slopes
Microcontroller: ESP32-S3 Dual-Core 240MHz	Web Visualizer: 3D React + Three.js Twin

PROJECT & TEAM SPECIFICATIONS (LaTeX Compiled Specification)

Project Title: GeoNail Geotechnical Monitoring Engine
Team Name: GeoNail Innovations
Team Lead: Yogarathinam
Development Team: Embedded Hardware, Signal Processing & 3D Web Twin Team
Repository: github.com/Yogarathinam/GeoNail_SIH26025
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Project & Team Information

1. Executive Summary

GeoNail is an advanced Internet of Subsurface Things (IoST) geotechnical monitoring platform designed for real-time detection of slope instability, structural micro-displacements, and seismic anomalies in high-risk mining, civil excavation, and landslide-prone environments.

The system integrates high-precision embedded sensor hardware, on-device edge anomaly processing, multi-transport telemetry (BLE, Serial UART, Wi-Fi/MQTT), and a 1:1 real-world scale 3D Web CAD & Digital Twin Visualizer built on Three.js and React.

+-----+ GEONAIL SYSTEM +-----+					
+-----+	+-----+		+-----+	+-----+	
Subsurface Anchors	M5Stack Edge Node		GeoNail OS Web		
- Magnetometer Array	-> - ESP32-S3 Dual-Core		-> - 3D CAD X-Ray		
- 3-Axis Accelerom.	- Compl. Filter / EMA		- 1:1 Minefield		
- Gyro Sensor	- Real-time Edge Rules		- BLE/Serial Modem		
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2. Hardware Architecture & Mechanical Design

2.1 Mechanical Form Factor

The GeoNail housing consists of a heavy-duty industrial-grade sealed polycarbonate shaft housing the electronics core, topped with a locked protective enclosure cap and anchored deep into soil/rock strata via a telescoping magnetic spike anchor.

- **Main Enclosure Body:** Cylindrical shell containing M5Stack Core2 / ESP32-S3 MCU, OLED/IPS display, lithium-polymer power cell, and status transducer (piezo/buzzer).
- **Telescoping Spike Anchor:** Underground anchor embedded directly into stable soil layers. As soil shifts, relative displacement between the outer casing and inner spike anchor alters internal magnetic field geometry.
- **Locked Top Cap:** Moisture-sealed protective cap housing external antennas and status LEDs.

2.2 Microcontroller & Sensors

- **Microcontroller:** ESP32-S3 (Dual-Core Xtensa LX7 @ 240 MHz, 512KB SRAM, 8MB PSRAM, Wi-Fi 4 + BLE 5.0).

- **Power Management Unit (PMU):** AXP192 integrated power controller supporting battery state-of-charge tracking, dynamic voltage scaling, and solar/external power charging.
- **Inertial Measurement Unit (IMU):** MPU6050 / MPU6886 6-Axis motion sensor (3-axis Accelerometer, 3-axis Gyroscope).
- **Magnetometer / Hall Sensor Array:** Precision magnetic sensing elements configured along the inner shaft axis to resolve absolute linear displacement of the subterranean anchor rod.

3. Firmware Architecture & Edge Signal Processing

The firmware is written in C++ for the Arduino/ESP32 framework (`GeoNail_M5Stack_Firmware.ino`). It executes multi-sensor sampling at 100 Hz with real-time digital filtering and edge anomaly detection.

3.1 Sensor Fusion & Orientation Math

To compute real-time Roll, Pitch, and Yaw angles without gimbal lock, the firmware implements a Complementary Filter combining raw gyroscope angular rates with accelerometer gravity vectors.

- **Accelerometer Roll Angle:**

$$\phi_{acc} = \arctan2(A_y, A_z) \cdot \frac{180}{\pi}$$

- **Accelerometer Pitch Angle:**

$$\theta_{acc} = \arctan2\left(-A_x, \sqrt{A_y^2 + A_z^2}\right) \cdot \frac{180}{\pi}$$

- **Complementary Filter Integration:**

$$\phi_t = \alpha \cdot (\phi_{t-1} + G_x \cdot \Delta t) + (1 - \alpha) \cdot \phi_{acc}$$

$$\theta_t = \alpha \cdot (\theta_{t-1} + G_y \cdot \Delta t) + (1 - \alpha) \cdot \theta_{acc}$$

$$\psi_t = \psi_{t-1} + G_z \cdot \Delta t$$

(where $\alpha = 0.96$ is the filter tuning constant, balancing gyroscope responsiveness with accelerometer low-pass drift suppression).

3.2 Subterranean Anchor Displacement Math

The magnetic flux density B measured along the longitudinal axis varies inverse-proportionally with the insertion distance d of the magnetic spike anchor relative to the casing baseline:

- **Magnetic Field Change:**

$$\Delta B = B_{ref} - B_{measured}$$

- **Linear Soil Displacement (mm):**

$$d_{mm} = k_m \cdot \sqrt{\frac{1}{\Delta B + \epsilon}} + d_{offset}$$

- As the spike anchor shifts deeper or pulls out during soil subsidence, ΔB changes dynamically.
- The calibrated constant k_m converts micro-tesla field variations into exact linear displacement in millimeters (mm) and centimeters (cm).

3.3 Edge Anomaly & Seismic Detection Engine

To eliminate environmental noise while guaranteeing immediate triggering during shear collapses or seismic shocks, the firmware utilizes dual statistical estimators:

- **Exponential Moving Average (EMA):**

$$EMA_t = \beta \cdot X_t + (1 - \beta) \cdot EMA_{t-1}$$

(where $\beta = 0.05$ smooths slow creep baseline drift).

- **Root Mean Square Acceleration (RMS_vib):**

$$RMS_{vib} = \sqrt{\frac{1}{N} \sum_{i=1}^N (A_{x,i}^2 + A_{y,i}^2 + A_{z,i}^2 - 1.0^2)}$$

Multi-Stage Trigger Thresholds:

	Sensor Condition	System Action & Telemetry		
		$\Delta Angle$	$< 2.0^\circ, d < 5.0 \text{ mm}, RMS < 0.15g$	Display regu
	$2.0^\circ \leq$	$\Delta Angle$	$< 5.0^\circ \text{ OR } 5.0 \text{ mm} \leq d < 15.0 \text{ mm}$	Amber LED prompt
		$\Delta Angle$	$\geq 5.0^\circ \text{ OR } d \geq 15.0 \text{ mm}$	Red LED str

	RMS_vib > 0.65g (Transient shock spike)	Instant emergency telemetry beacon dispatch		
	Rate of displacement $\partial d/\partial t > 2.5$ mm/s	Lock memory log dump, trigger shear collapse alert		

4. Telemetry Protocols & Communications

GeoNail provides continuous multi-channel telemetry streams supporting three distinct operational modes:

4.1 USB Serial UART Telemetry (115200 Baud)

Sends structured JSON packets at 20 Hz over the USB-C serial port:

```
{
  "device_id": "GEONAIL_NODE_01",
  "uptime_ms": 142850,
  "pitch": 1.45,
  "roll": -0.82,
  "yaw": 12.30,
  "displacement_mm": 3.42,
  "vibration_rms": 0.042,
  "battery_pct": 94,
  "status": "NORMAL",
  "rssi": -62
}
```

4.2 Bluetooth Low Energy (BLE) GATT Architecture

- **Device Name:** GeoNail-Node-01
- **Custom Service UUID:** 4FA10001-0000-1000-8000-00805F9B34FB
- **Characteristics:**
 - **Telemetry Data (Notify):** 4FA10002-0000-1000-8000-00805F9B34FB (Binary packed float array: [Pitch, Roll, Yaw, Displacement, Vibration]).
 - **Calibration Command (Write):** 4FA10003-0000-1000-8000-00805F9B34FB (Send 0x01 to zero IMU tilt, 0x02 to reset magnet baseline).

4.3 Wi-Fi Mesh & MQTT WebSockets

In remote deployments, nodes form a local ESP-NOW mesh relaying telemetry to a central gateway node, which publishes MQTT topics (`geonail/telemetry/node_01`) over cellular or satellite backhaul to the GeoNail OS visualizer.

5. GeoNail OS Web Visualizer Architecture

The frontend visualizer is a modern React 18 + TypeScript web application (**sih-3d-visualizer**) powered by Vite, Tailwind CSS, Lucide icons, and Three.js / React Three Fiber.

5.1 System Tab Architecture

GEONAIL OS INTERFACE			
[1. 3D CAD Viewer]	[2. MCU Dashboard]	[3. Minefield Digital Twin]	
- High-res geometry	- Real-time gauges	- 1:1 scale (800m x 600m)	
- X-Ray translucent	- Live trend graphs	- CAT 797 heavy dump trucks	
- Locked cap state	- Direct Serial/BLE	- Subterranean shear slice	
- Multi-camera system	- Alarm logs & reset	- Node deployment grid	

5.2 3D CAD Viewer Page Features

- **Full 3-Axis Reactive Model:** Real-time rotational synchronization driven by live or simulated Pitch, Roll, and Yaw angles.
- **Translucent X-Ray Mode:** Renders exterior casing with glass-like transparency (opacity = 0.35), exposing internal battery, M5Stack core board, magnetometer coils, and internal anchoring mechanism.
- **White Engineering Grid Aesthetics:** Crisp white grid floor background, slate grid lines, and axis markers for clear CAD visualization.
- **Locked Cap State:** Standardized fixed state showing locked protective cap securely attached to top shaft.
- **Multi-Preset Camera Control:** Dedicated controls for standard engineering projections:
- **Isometric View:** Position [12, 10, 12], Target [0, 0, 0]
- **Top (Plan) View:** Position [0, 22, 0], Target [0, 0, 0]
- **Front Elevation:** Position [0, 0, 20], Target [0, 0, 0]
- **Side Elevation:** Position [20, 0, 0], Target [0, 0, 0]
- **Reset Orbit:** Smoothed reset transitions via OrbitControls.

5.3 MCU Dashboard Page Features

- **Hardware Connection Modal:** Native Web Serial API & Web Bluetooth API integration allowing operators to connect directly to physical GeoNail devices from any browser.
- **Interactive Telemetry Gauges:** Real-time dial gauges for Pitch (°), Roll (°), Displacement (mm), and Vibration RMS (g).
- **Historical Signal Trends:** Canvas-based line graphs tracking 60-second sliding windows of ground movement.
- **System State Indicator:** Dynamic color-coded status banner with audio warning indicators.

5.4 1:1 Scale Minefield Digital Twin Page Features

- **Real-World Environment:** Models an 800m × 600m open-cast mining pit complete with terraced benches (5m height steps), haulage ramps, excavation zones, and top perimeter access roads.
 - **Real-World Scale Reference:** Features detailed 1:1 scale 3D models of **CAT 797 Dump Trucks** (14.8m × 9.8m × 6.5m) navigating haul roads, allowing engineers to intuitively judge scale.
 - **Multi-Node Deployment Grid:** Multi-node monitoring network (**GEONAIL-01** to **GEONAIL-06**) positioned along critical slope benches. Clicking any node selects it, displaying its live telemetry HUD overlay and triggering a smooth camera glide to its coordinates.
 - **Subterranean Shear Slice Visualization:** Underground cross-section plane highlighting slip surfaces, failure arcs, soil strata layers, and real-time anchor penetration depths.
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6. Calibration & Operating Procedures

6.1 Initial Zero-Bias Calibration Procedure

1. **Mechanical Mounting:** Drive GeoNail outer casing into borehole until top flange rests flush against reference surface.
2. **Spike Extension:** Drive telescoping spike anchor into bedrock / solid substrate.
3. **Firmware Zeroing:**
 - Hold M5Stack **Button A** for 3 seconds or trigger **Zero Calibration** in GeoNail OS Web Visualizer.
 - Device records current A_x , A_y , A_z baseline offset as 0.0° pitch/roll reference.
 - Device stores current magnetic flux reading B_0 as zero displacement (0.0 mm).

6.2 Troubleshooting Audio & Display Behavior

- **Audio Noise / Buzzing:** Occurs when PWM buzzer pin is left floating during WiFi transmission bursts. Solved in firmware v2.1 via active pull-down logic and low-pass filter capacitor across speaker lines.
 - **Display Flicker:** Prevents display burn-in and SPI bus contention by utilizing double-buffered sprite rendering (`M5.Lcd.pushSprite()`) updated strictly at 30 FPS.
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7. System Verification & Validation

The full GeoNail system architecture has undergone extensive end-to-end integration testing:

1. **Hardware Communication:** Web Serial and BLE hardware communication modules verified at 115200 baud and 20 Hz packet rates.
2. **Orientation Accuracy:** Roll/Pitch tracking verified within $\pm 0.2^\circ$ accuracy against physical digital clinometer.

- 3. Anchor Displacement Resolution:** Magnetometer displacement calculation validated from 0.0 mm to 50.0 mm with sub-millimeter precision.
 - 4. Digital Twin Visualization:** Smooth 60 FPS 3D canvas rendering verified across CAD, MCU, and Minefield views.
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Documentation compiled for GeoNail Open-Cast Mining & Subterranean Geotechnical Monitoring System (SIH 2026).